

Saurabh Annadate

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Resume Summary

Atmospheric scientist with a PhD in emissions estimation using inverse modelling. Experienced in quantifying greenhouse gas and aerosol emissions with transport models like FLEXPART and WRF. Proficient in Python, Fortran, and high-performance computing, with strong expertise in climate data analysis and machine learning. Contributed to EU HORIZON projects like PARIS and EYE-CLIMA, delivering policy-relevant insights. Driven by a passion for applying data-driven methods to address climate challenges and support informed decision-making.

Education

PhD in Climate Change and Sustainable Development from Scuola Superiore Studi Pavia IUSS, Italy Nov 21 - Oct 24

- **Thesis:** Emission estimates of radiatively active species at the regional scale using inverse modelling techniques. [link](#) [🔗](#)

Bachelor and Master of Science (BS-MS) dual degree from Indian Institute of Science Education and Research Mohali, India Aug 16 - May 21

- Major in Physics and minor in Data Science
- **Cumulative Performance Index (CPI):** 8.23/10
- **Thesis:** A modelling study using WRF-Chem over north India with improved transport and waste burning emissions. [link](#) [🔗](#)

Honors and Awards

- **Outgoing Visiting Researchers Grant Award 2025** — Awarded by the University of Urbino to conduct research activities at the University of Vienna.
- **Doctor Europaeus Distinction** — Conferred alongside the PhD, recognizing the European dimension of the doctoral degree.
- **ERASMUS+ Mobility Grant 2023** — Awarded by the European Union for research traineeship abroad.
- **Prime Minister's Research Fellowship (PMRF)** — Prestigious fellowship awarded by the Ministry of Education, Government of India.
- **INSPIRE Scholarship (SEATS)** — Awarded under the Innovation in Science Pursuit for Inspired Research scheme by the Department of Science and Technology (DST), Government of India.
- **Merit-cum-Means Fellowship, IISER Mohali** — Granted for academic excellence and financial need.

Skills

Languages: Python, Fortran, Bash, UNIX, git, L^AT_EX

Atmospheric modelling: FLEXPART, FLEXINVERT+, WRF, WRF-Chem

Tools: CDO, NCO, xarray, Panoply, openMP

HPC: Experienced in using HPC clusters such as Leonardo, Cheyenne.

Measurements: Hands-on experience with trace-gas analysers (CO, NO_x, O₃).

Machine learning: Applied experience with [neural networks](#) [🔗](#), [natural language processing](#) [🔗](#), and SVMs.

Experience

Post-doctoral researcher *University of Urbino*

Urbino, IT
Nov 2024 – Current

- Leading the deliverable that aims to estimate European black carbon emission fluxes using top-down methods under the [PARIS](#)  project.
- Developing an optimization algorithm for wet-scavenging coefficients in the wet deposition parametrization scheme in the FLEXPART model.

PhD thesis *IUSS Pavia*

Pavia, IT
Nov 21 - Oct 24

- Optimised an inversion framework (FLEXINVERT) to estimate F-gas emissions over Europe and Italy.
- Estimated impact of the European regulation on long-term trends of Italian HFC-134a emissions.
- Collaborated with national inventory makers to reduce the emissions gap between top-down and bottom-up estimates.

Masters thesis *IISER Mohali*

Mohali, IN
Aug 20 - May 21

- Incorporated new road transport and waste-burning emission inventories in the WRF-Chem setup.
- Estimated discrepancies in road transport and waste burning emissions.
- Studied effects on local and regional air quality.

Publications

1. ANNADATE, S., MANCINELLI, E., GONELLA, B., MORICCI, F., O'DOHERTY, S., STANLEY, K., YOUNG, D., VOLLMER, M. K., CESARI, R., FALASCA, S., ET AL. Monitoring the impact of eu f-gas regulation on hfc-134a emissions through a comparison of top-down and bottom-up estimates. *Environmental Sciences Europe* 37, 1 (2025), 1–12
2. HUNDAL, R. A., ANNADATE, S., CESARI, R., COLLALTI, A., MAIONE, M., AND CRISTOFANELLI, P. Emissions of climate-altering species from open vegetation fires in the Mediterranean region - A review on methods and data. *Science of The Total Environment* 957 (2024), 177713
3. VOJTA, M., PLACH, A., ANNADATE, S., PARK, S., LEE, G., PUROHIT, P., LINDL, F., LAN, X., MÜHLE, J., THOMPSON, R. L., AND STOHL, A. A global re-analysis of regionally resolved emissions and atmospheric mole fractions of SF₆ for the period 2005–2021. *Atmospheric Chemistry and Physics* 24, 21 (2024), 12465–12493. Publisher: Copernicus GmbH
4. ANNADATE, S., FALASCA, S., CESARI, R., GIOSTRA, U., MAIONE, M., AND ARDUINI, J. A Sensitivity Study of a Bayesian Inversion Model Used to Estimate Emissions of Synthetic Greenhouse Gases at the European Scale. *Atmosphere* 15, 1 (2024), 51. Number: 1 Publisher: Multidisciplinary Digital Publishing Institute
5. SHARMA, G., ANNADATE, S., AND SINHA, B. Will open waste burning become India's largest air pollution source? *Environmental Pollution* 292 (2022), 118310. Publisher: Elsevier
6. HAKKIM, H., KUMAR, A., ANNADATE, S., SINHA, B., AND SINHA, V. RTEII: A new high-resolution (0.1° × 0.1°) road transport emission inventory for India of 74 speciated NMVOCs, CO, NO_x, NH₃, CH₄, CO₂, PM_{2.5} reveals massive overestimation of NO_x and CO and missing nitromethane emissions by existing inventories. *Atmospheric Environment: X* 11 (Oct. 2021), 100118

Conferences and Workshops

1. European Aerosol Conference (EAC-2025), Lecce, Italy (Oral presentation) *June 2025*
 - **Title:** Top-down estimates of European emissions of black carbon for 2022.
2. 21st International Workshop on Greenhouse Gas Measurements from Space (IWGGMS-21), Takamatsu, Japan (poster presentation) *June 2025*
 - **Title:** Top-down estimates of European emissions of black carbon for 2022.

3. European Geosciences Union (**EGU**) General Assembly (oral presentation) *April 2024*
 - **Title:** Estimates of HFC-134a Emissions over Europe informed by observations show a recent increase.
4. PARIS-Horizon EU project Annual Meeting, Bristol, UK (oral presentation) *Jan 2024*
 - **Title:** Top-down Italian emissions of HFC-134a: 15 years trend and a comparison with the Italian national inventory reports.
5. GEIA: Global Emissions Initiative 20th conference, Brussels, Belgium (poster) *June 2023*
 - **Title:** Long-term emission estimates of radiatively active species at the regional scale using inverse modelling techniques.
6. DOSC 2025: Debugging and Optimization of Scientific Applications - CINECA, Bologna, Italy. [certificate](#) [🔗](#) *March 2025*
7. AMIGO: Atmospheric chemistry modeling, data assimilation, inverse modeling, and model evaluation. BIRA-IASB, Belgium. *June 2023*
8. DARE UK and Class International Summer School 2022, Southampton, UK *July 2022*
[certificate](#) [🔗](#).

Teaching Experience

- **Climate Modelling:** Conducted a practical session on downloading and analyzing CMIP6 data in the climate modelling course at the University of Urbino. [link](#) [🔗](#)
- **Introduction to Atmospheric Science:** Served as a teaching assistant, supporting course delivery and student learning at IISER Mohali.